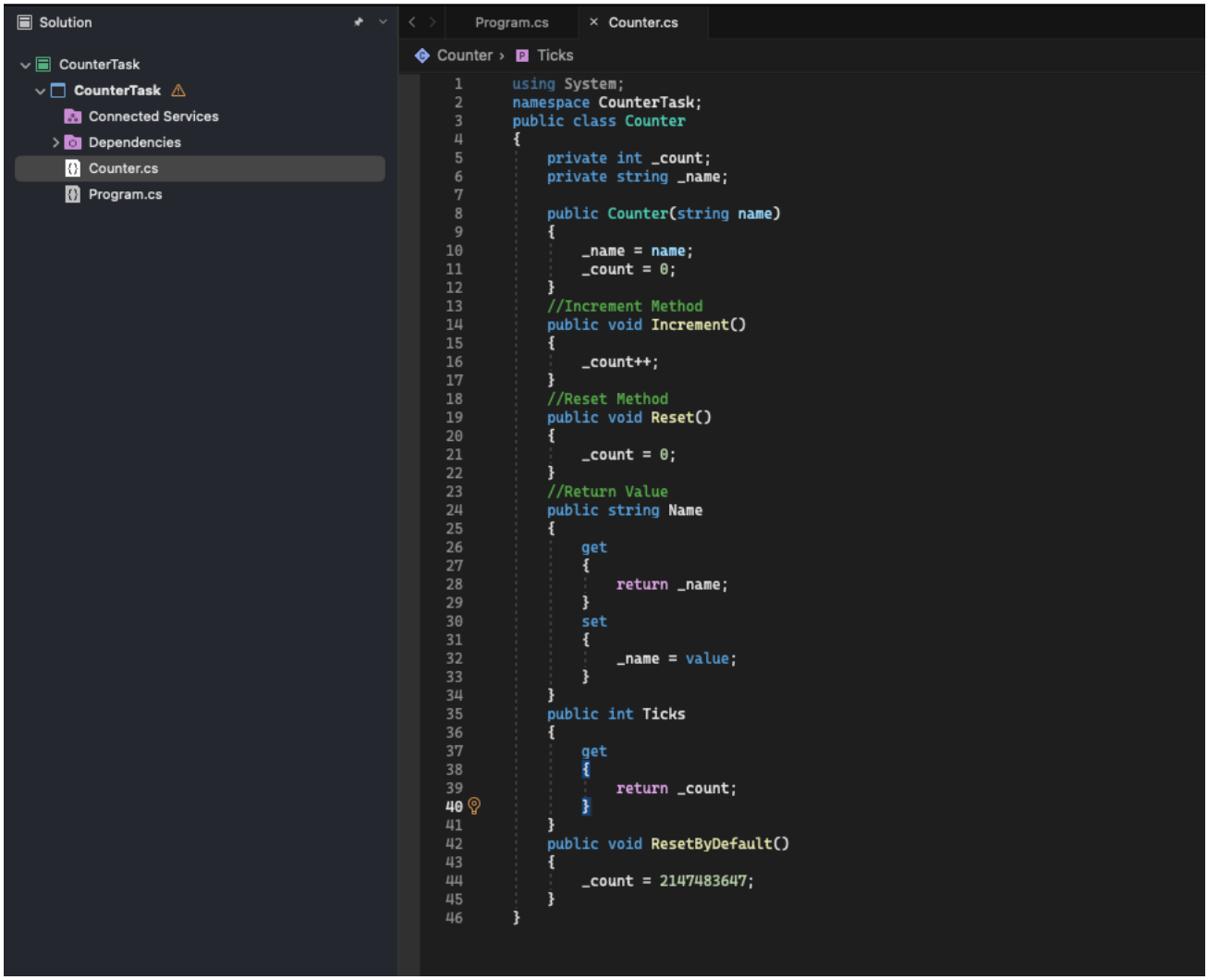
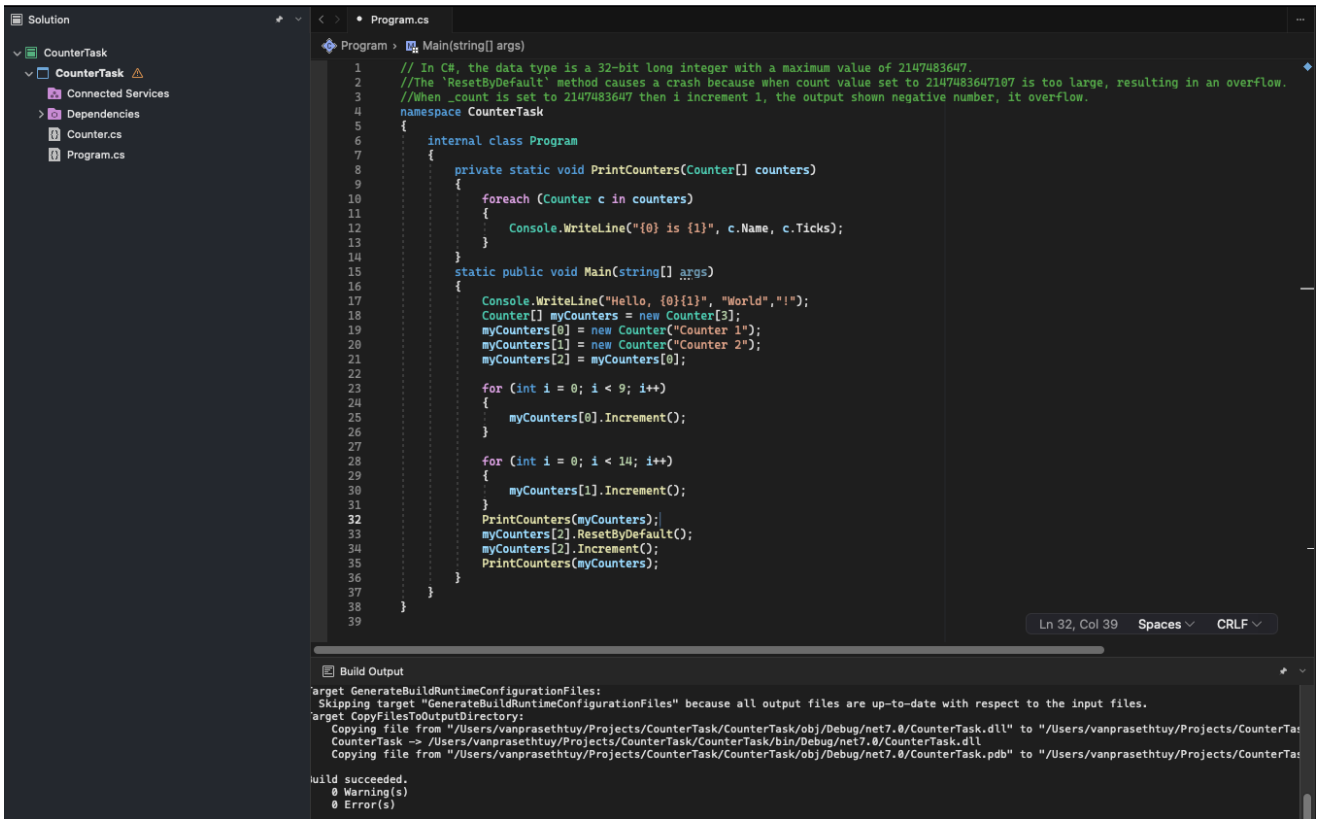
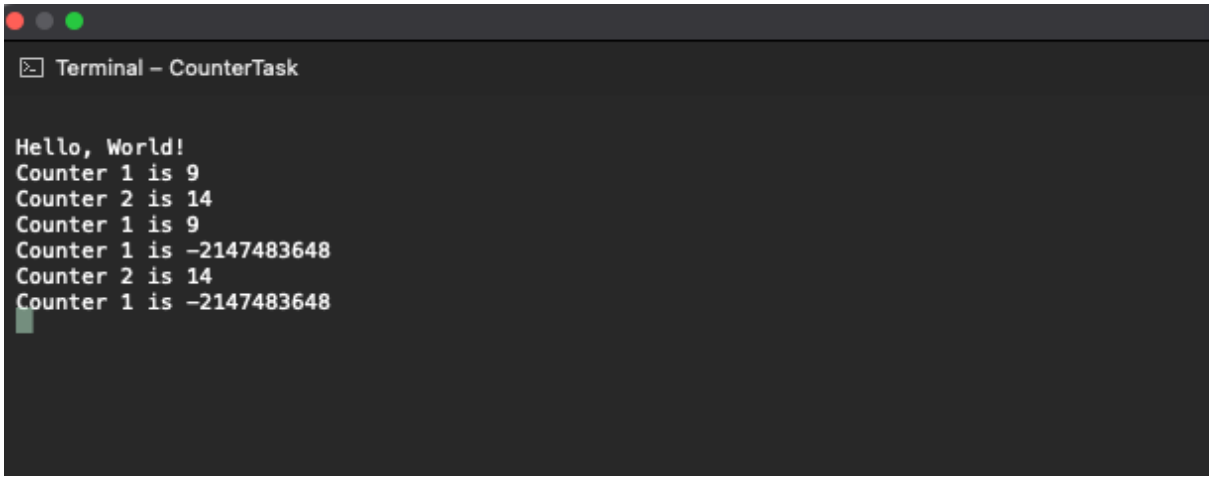
Pass Task 2.2: Counter Class and Arithmetic Overflow-checking

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Solution
CounterTask
CounterTask
Connected Services
Dependencies
Counter.cs
Program.cs
Program.cs
Program › Main(string[] args)
// In C#, the data type is a 32-bit long integer with a maximum value of 2147483647.
//The `ResetByDefault` method causes a crash because when count value set to 2147483647107 is too large, resulting in an overflow.
//When _count is set to 2147483647 then i increment 1, the output shown negative number, it overflow.
namespace CounterTask
{
internal class Program
{
private static void PrintCounters(Counter[] counters)
{
foreach (Counter c in counters)
{
Console.WriteLine("{0} is {1}", c.Name, c.Ticks);
}
}
static public void Main(string[] args)
{
Console.WriteLine("Hello, {0}{1}", "World","!");
Counter[] myCounters = new Counter[3];
myCounters[0] = new Counter("Counter 1");
myCounters[1] = new Counter("Counter 2");
myCounters[2] = myCounters[0];
for (int i = 0; i < 9; i++)
{
myCounters[0].Increment();
}
for (int i = 0; i < 14; i++)
{
myCounters[1].Increment();
}
PrintCounters(myCounters);
myCounters[2].ResetByDefault();
myCounters[2].Increment();
PrintCounters(myCounters);
}
}
}
Ln 32, Col 39
Spaces
CRLF
Build Output
arget GenerateBuildRuntimeConfigurationFiles:
Skipping target "GenerateBuildRuntimeConfigurationFiles" because all output files are up-to-date with respect to the input files.
arget CopyFilesToOutputDirectory:
Copying file from "/Users/vanprasethtuy/Projects/CounterTask/CounterTask/obj/Debug/net7.0/CounterTask.dll" to "/Users/vanprasethtuy/Projects/CounterTas
CounterTask -> /Users/vanprasethtuy/Projects/CounterTask/CounterTask/bin/Debug/net7.0/CounterTask.dll
Copying file from "/Users/vanprasethtuy/Projects/CounterTask/CounterTask/obj/Debug/net7.0/CounterTask.pdb" to "/Users/vanprasethtuy/Projects/CounterTas
uild succeeded.
0 Warning(s)
0 Error(s)


Terminal – CounterTask
Hello, World!
Counter 1 is 9
Counter 2 is 14
Counter 1 is 9
Counter 1 is -2147483648
Counter 2 is 14
Counter 1 is -2147483648